

From cells to donor-level prediction

One CMV label per donor

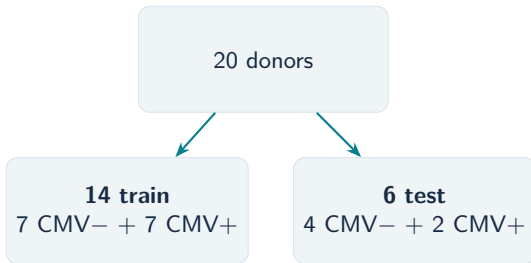
Donors	20
CMV- / +	11 / 9
Protein markers	37
Input gate	<code>gated_alive</code>

Live, singlet PBMCs; no extra NK gate.

$$a_{ij} = \text{asinh}(x_{ij}/5)$$

CellCNN/SVM: fit marker scaling on training donors only. Citrus uses ArcSinh values directly.

30 shared donor splits



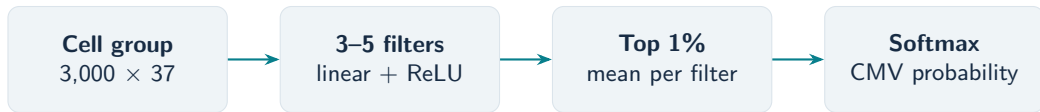
3 inner donor folds
for model selection

Evaluate once
on unseen donors

The independent observation is the donor. Cells from one donor stay together.

Horowitz et al. (2013), [doi:10.1126/scitranslmed.3006702](https://doi.org/10.1126/scitranslmed.3006702). Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

CellCNN learns filters for informative cells



Learn which cells matter

$$r_{if} = \max(0, w_f^\top z_i + b_f)$$

A filter scores a marker combination. Pooling focuses on the strongest responses; the whole network learns from donor labels.

Our paper-guided implementation

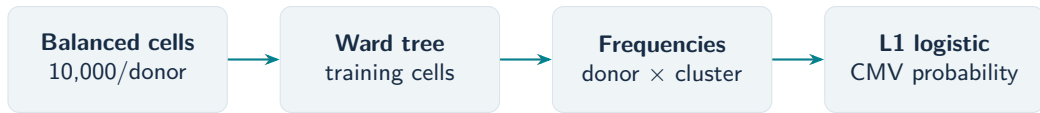
- PyTorch; 200 cell groups per training donor.
- Choose the best inner-fold network; no outer refit.
- Adaptations: fixed 3/4/5-filter search; average five test-input probabilities.

Filter learning and donor classification are trained jointly.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

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Citrus predicts from population frequencies



First define populations

Retain tree clusters containing at least **0.05% of training cells**. Parent and child clusters may overlap.

Each donor is represented by the fraction of its cells in each retained cluster.

Then select predictive frequencies

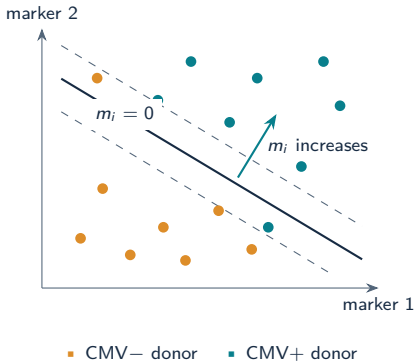
Nonzero L1-logistic coefficients identify selected model populations. New cells inherit memberships from their nearest training cell.

Compute compromise: 10,000 cells and 30 splits, versus 20,000 and 100 in the paper.

Citrus separates population discovery from supervised prediction.

Bruggner et al. (2014), [doi:10.1073/pnas.1408792111](https://doi.org/10.1073/pnas.1408792111). Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

A linear SVM assigns a score to each cell



Schematic in 2D; actual model uses 37 markers.

Train on donor-labelled cells

10,000 cells per training donor; each cell receives its donor's label.

$$m_i = w^\top z_i + b$$

A larger margin means stronger evidence in the model's positive direction.

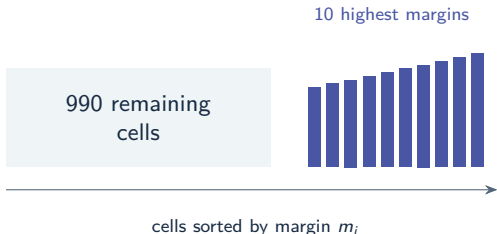
Linear SVM: L2 penalty and squared-hinge loss. The margin is a score, not a probability.

A donor label is weak supervision: it does not establish the state of each cell.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

Our adaptation: aggregate the highest cell margins

A few high-scoring cells can matter



Toy example: 1,000 cells; keep 10.

Margins 2.1, 2.2, ..., 3.0 give $s_d = 2.55$.

At $\tau = 1.80$: predict CMV+.

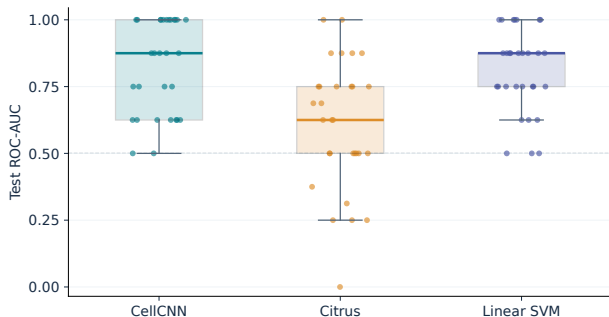
One score per donor

$$k_d = \lceil 0.01 N_d \rceil, \quad s_d = \frac{1}{k_d} \sum_{i \in \text{Top}_{k_d}} m_i$$

1. Select $C \in \{0.01, 0.1, 1\}$ by inner donor ROC-AUC.
2. Learn τ from inner out-of-fold donor scores.
3. Refit on 14 training donors; score all test cells.

Our project adaptation is the Top-1-% donor score, fixed before testing.

Performance across 30 shared donor splits



Median across 30 splits

Method	AUC	AP	BA
CellCNN	0.875	0.833	0.625
Citrus	0.625	0.542	0.500
Linear SVM	0.875	0.833	0.625

AUC: ranking of CMV+ versus CMV-.

AP: average precision.

BA: balanced accuracy.

Six test donors per split. Boxes show Q1–Q3; dots show individual splits.

CellCNN and SVM share the highest median AUC; split variability remains substantial.

Only 20 independent donors. Overlapping splits do not establish general superiority.

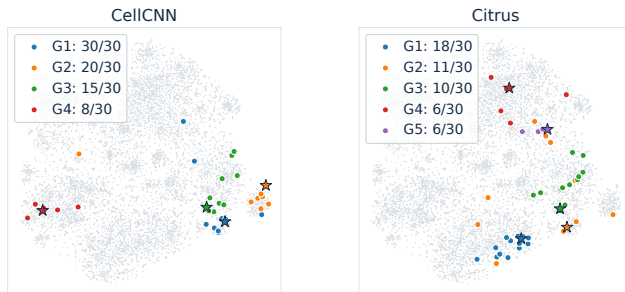
BA thresholds: 0.5 for CellCNN/Citrus; inner-trained for SVM. Quartiles are not confidence intervals.

Which CellCNN and Citrus subsets recur?

CellCNN: $r_{if} / \max_{j \in T_s} r_{jf} > 0.5$ on the original training reference.

Citrus: cluster centroids with nonzero logistic coefficients ($|\beta| > 10^{-10}$).

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.



$F_G = \#\{\text{splits containing group } G\}/30$; show $F_G \geq 20\%$. Top groups: **30/30** and **18/30**, both positive.

Projected training centroids; stars: representatives; group IDs/colours are method-specific.

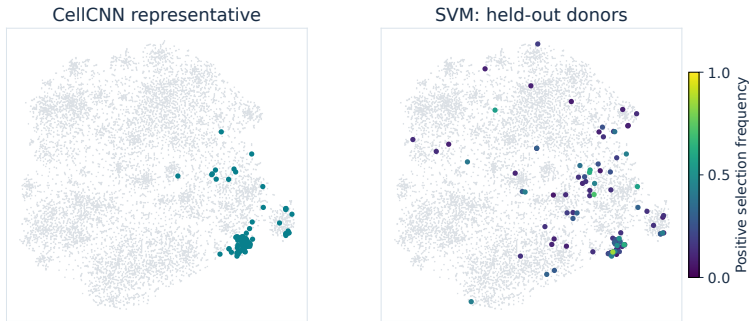
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Which cells does the SVM repeatedly select?

SVM: keep the highest 1% of margins in each full test donor; positive selection requires $m_i > \tau_s$.

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.



CellCNN: 87 map cells (exploratory). **SVM:** 93 map cells selected positively at least once.

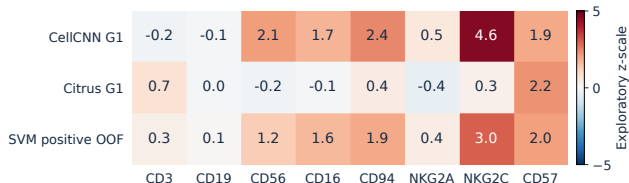
SVM denominator: all 4–16 test appearances of that cell's donor.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

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What do the marker profiles support?

CellCNN/Citrus: group representatives. SVM: positive test-cell profiles, with equal donor weights.



CellCNN

NKG2C/CD57 with CD56/CD16/CD94: compatible with a memory-like NK profile.

Citrus

Higher CD3/CD57 and much less NKG2C elevation suggest a T-associated pattern.

SVM positive OOF

NKG2C/CD57 and NK-associated markers are elevated; CD3 is also above the reference mean.

SVM: 168/180 nonempty test visits; 20/20 donors. Shared exploratory z-scale; different subset definitions.

These profiles support biological hypotheses, not validated cell-type assignments.

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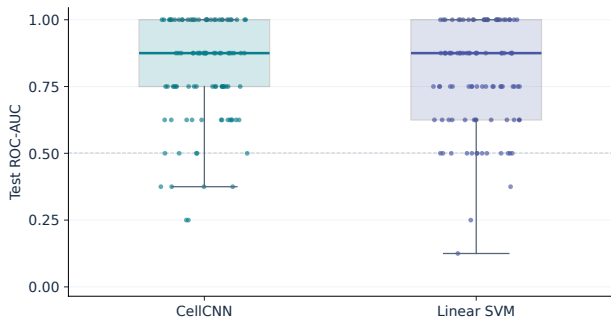
Backup: hyperparameters and paper deviations

	Implemented settings	Selection / paper relationship
CellCNN	3,000 cells/input; 200 inputs/donor; 3/4/5 filters; top 1%; Adam 0.01; batch 128; L2 10^{-4} ; no dropout; ≤ 100 epochs; patience 5.	Best validation accuracy; ties: AUC, loss, fewer filters, fold order. Fixed search; five rather than one test input. Selected net uses 9/10 donors.
Citrus	10,000 cells/donor; Ward; 37 ArcSinh markers; minimum cluster fraction 0.0005; L1 logistic; cv.min; no marker z-scoring.	10,000 cells/30 repeats versus paper's 20,000/100. Inner trees fitted per fold; original lambda grid uses all 14 outer training donors. Final fit on 14 donors.
SVM	10,000 cells/training donor; train-only scaler; L2 squared hinge; $C \in \{0.01, 0.1, 1\}$; top $\lceil 0.01N_d \rceil$ full-test margins.	Maximize mean inner AUC; ties: smaller C. Inner OOF threshold maximizes sensitivity + specificity. Final fit on 14 donors. Top-1-% aggregation is our adaptation.

Interpretation adaptation: average-linkage/cosine/cut 0.4 transferred from filter grouping to subset centroids.

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Backup: CellCNN and SVM across 100 splits



Separate two-method comparison

Method	AUC	AP	BA
CellCNN	0.875	0.833	0.750
Linear SVM	0.875	0.833	0.625

Medians across splits 0–99.

AUC: ROC-AUC; AP: average precision;
BA: balanced accuracy.

600 held-out donor predictions per method, from the same 20 donors.

Both median AUCs remain 0.875. More splits do not add independent donors.

Boxes: Q1–Q3; dots: individual splits; dashed line: AUC 0.5.